

Solar Charge Controllers



Solar Charger Controllers

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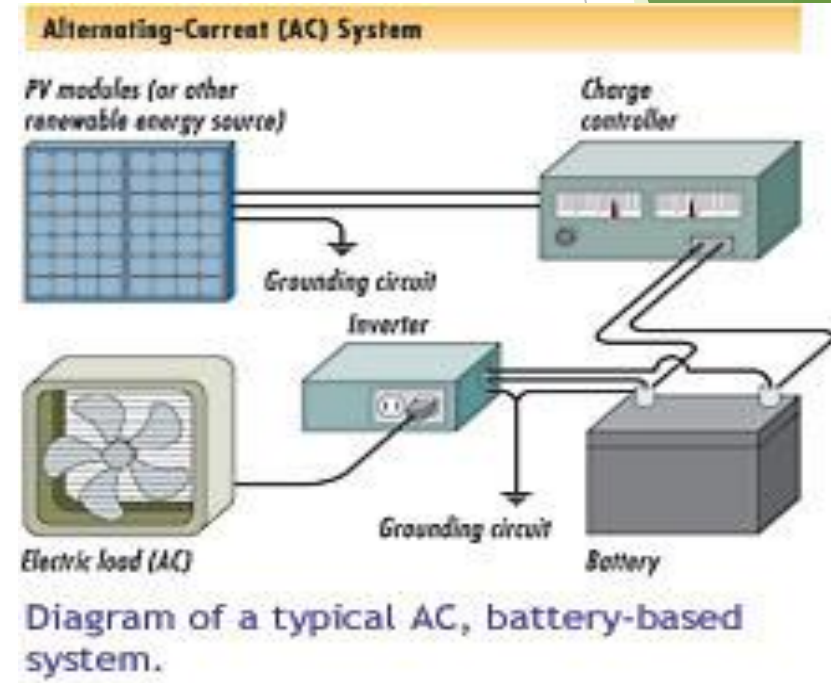
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Basic Functions

- Prevents battery overcharge
- Blocks reverse current
- Prevent over-discharge from batteries
(some models)
- Prevent electrical overload (some models)
- Displays battery status



Why have a Charge Controller ?

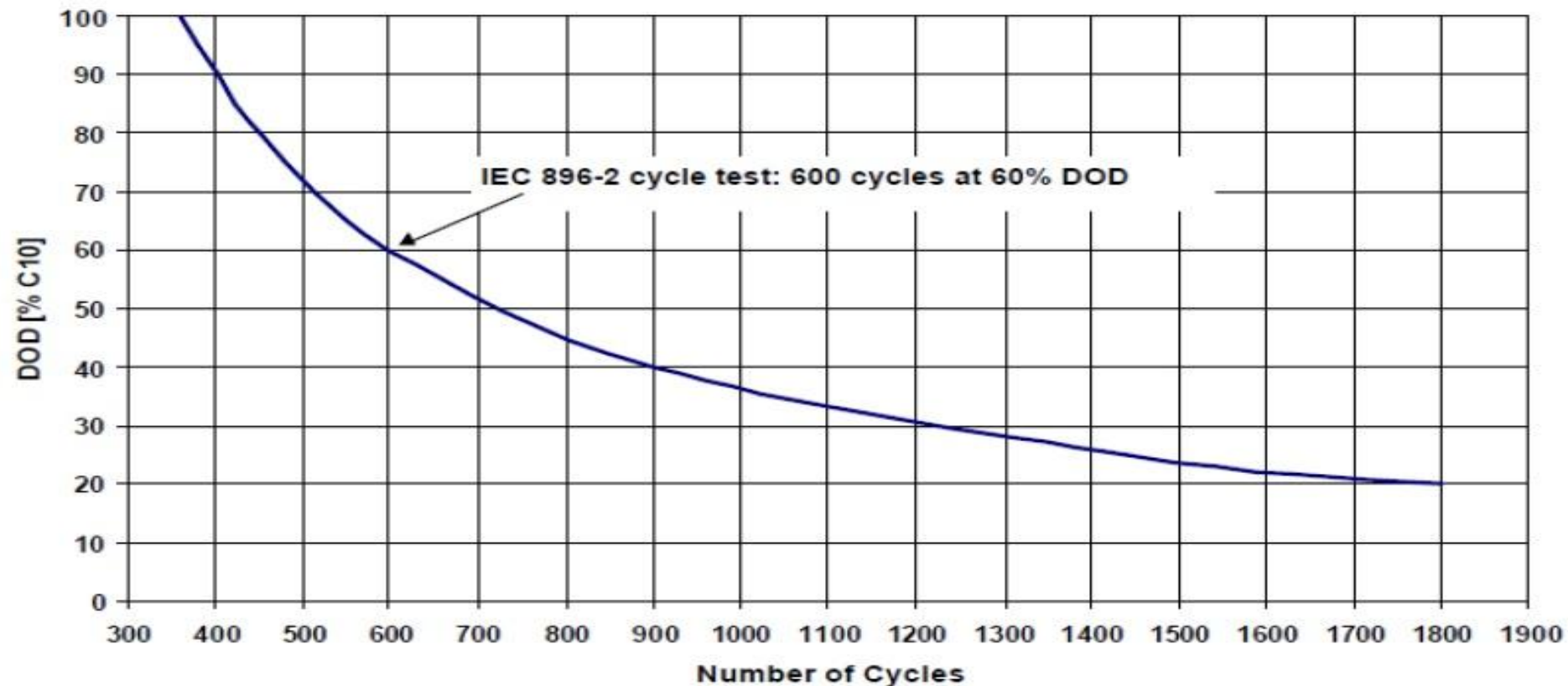
- ▶ Open cell batteries can lose water quickly if overcharged. If electrolyte falls below the top of the plates damage occurs.
- ▶ Sealed batteries may be ruined if frequently overcharged

Why have a Discharge Control?

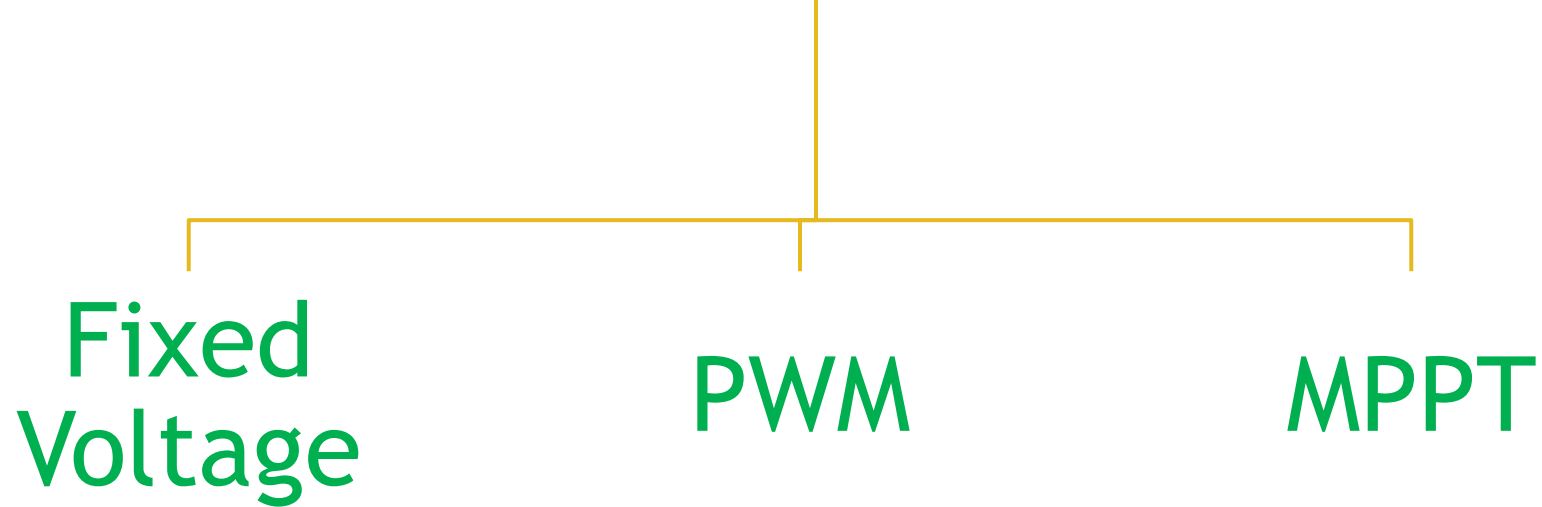
- ▶ To prevent damage to batteries from excessively deep discharge

Solar Batteries Lifetime Vs. DoD

Lifetime is significantly affected by the designed Depth of discharge (DOD)



Charge Controllers Types



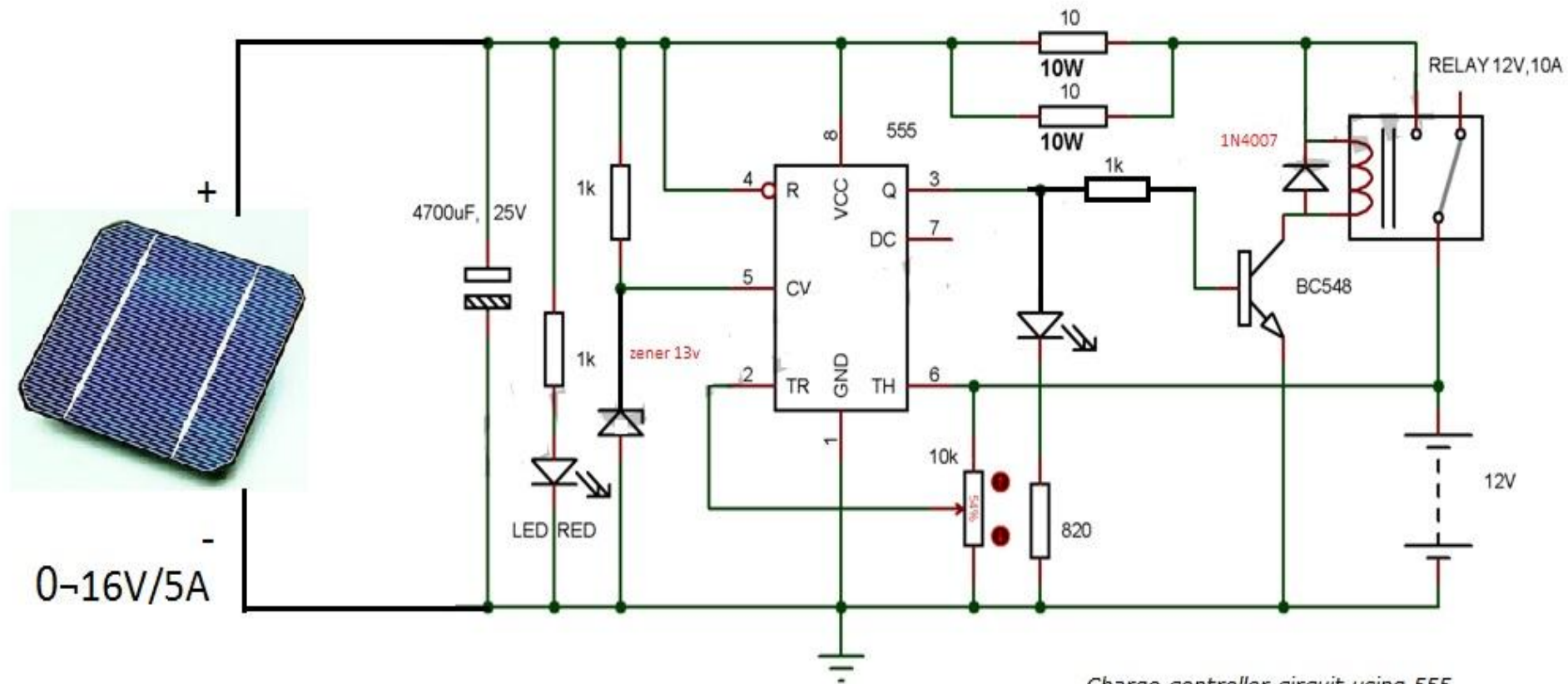
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graph TD; A[Charge Controllers Types] --> B[Fixed Voltage]; A --> C[PWM]; A --> D[MPPT]
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Fixed
Voltage

PWM

MPPT

Fixed Voltage Charger



Charge controller circuit using 555

Advantages

- ▶ Simple and cheap. Lends itself to local production
- ▶ Less likely to be damaged by excess current flow than series switching control.

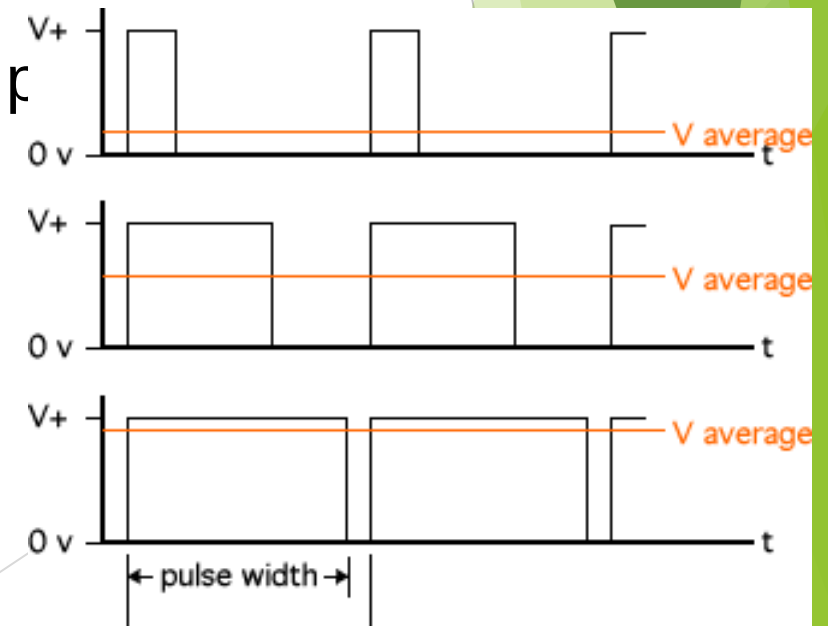
Disadvantages

- ▶ All the panel power is shorted through the controller and converted to heat. So large panels generate a great deal of heat and it is a problem to get rid of it. Shunt controls are therefore best suited to small (under 50Wp) PV systems.
- ▶ Sensitive to lightning
- ▶ Requires a blocking diode with its attendant power loss

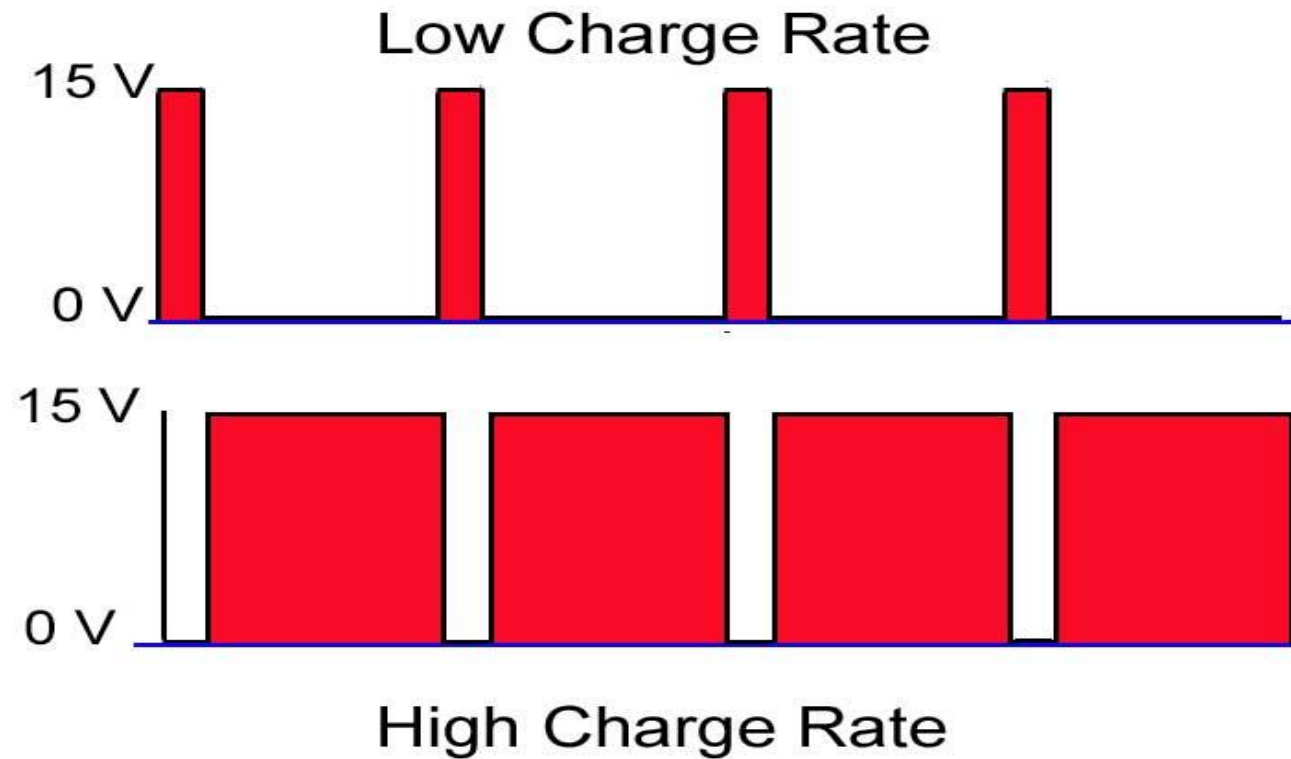
Pulse-Width Modulation (PWM) Controllers

Battery is charged using pulses of energy that can vary in length and speed according to battery condition.

- Ideal for situations where excess energy is often present
 - higher charging efficiency
 - rapid recharging
- healthy battery at full capacity



Changing PWM charging rate



Red area represents energy going to battery for charging

PWM Charger

helping to maximize battery life



Maximum Power Point Tracking (MPPT) Controllers

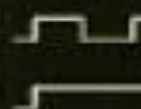
- A MPPT controller adjusts the voltage output to take advantage of the V_{pp} and charge the battery more .
- *Peak Power Voltage* (V_{pp}) is the maximum power point that a PV system can deliver; varies with temperature and sunlight intensity.
- V_{pp} is higher in colder and sunnier conditions therefore, MPPT controllers are ideal for cold climate use where battery use is high

SOLAR BOOST 2000E

25 AMP MAXIMUM POWER POINT TRACKING SOLAR CHARGE CONTROLLER

12.66

CHARGE STATUS



CHARGED



CHARGING



EQUALIZE

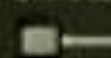


BATTERY VOLTAGE

SOLAR PANEL CURRENT

OUTPUT CHARGE CURRENT

CHARGE MODE



NORMAL



EQUALIZE



NOT ALL BATTERIES CAN BE SAFELY EQUALIZED.
FOLLOW BATTERY MANUFACTURER'S RECOMMENDATIONS
REGARDING TO EQUALIZATION. SEE OPERATOR MANUAL.

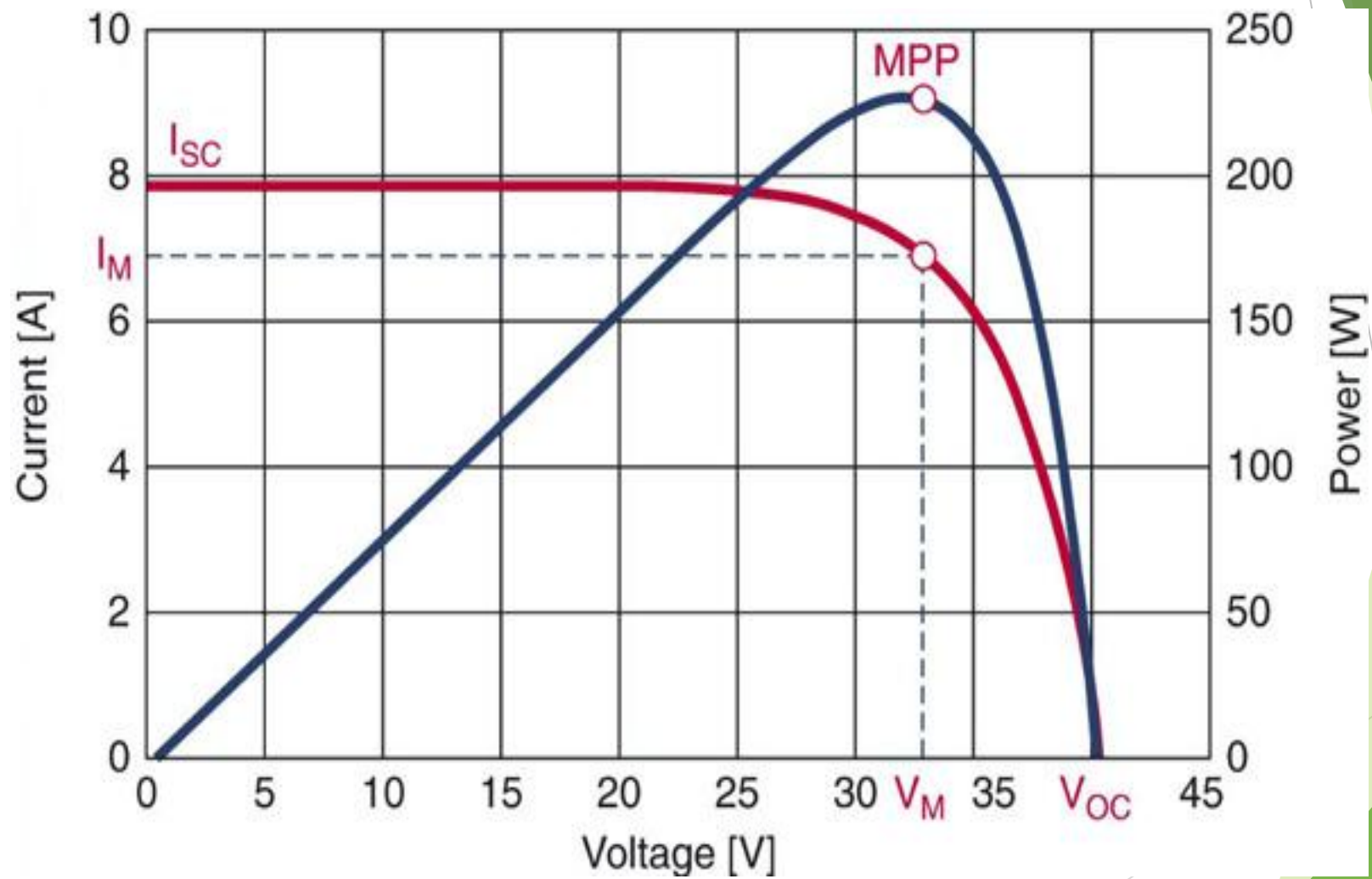
RV POWER PRODUCTS

ENCINITAS, CA USA

Sample MPPT controller



- ▶ Outback MX-60 MPPT controller
- ▶ Includes data logger and remote monitoring
- ▶ Programmable equalization
- ▶ Input voltage up to 140 DC to charge a battery ranging from 12V-60V
- ▶ Claims up to 30% more energy than non MPPT controllers but more likely 10%-15% if panel voltage is matched to that of batteries.



Discharge controller

- ▶ Placed between the battery and the loads
- ▶ All are of the series switching type. Often use a relay even if the charge controller uses semiconductors but many use a semiconductor switch

Charge Controllers: MPPT vs. PWM

PWM Charge Control

- ▶ PV Array and Battery Voltages must match.
- ▶ Array is sized in Amps based in I_{mp} of modules.
- ▶ Less expensive
- ▶ Simpler series switching charge control circuit using PWM pulse charge control.

MPPT Charge Control

- ▶ Array can be higher nominal voltage than battery and controller converts power down.
- ▶ Array is sized in watts
- ▶ High Frequency charge control